

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Deepak R (192511008) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

① Definition

Cloud computing is a model that enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources such as servers, storage, networks, applications and services. These resources can be rapidly provisioned and released with minimal management effort or service provider interaction. Instead of purchasing and maintaining physical hardware users can access computing resources over the Internet and pay only for the services they use. This pay-per-use model ~~for~~ makes cloud computing cost effective, flexible and highly scalable.

Characteristics

① On demand Self-Service

In traditional systems, purchasing and configuring hardware hardware may take several days or weeks whereas cloud resources are available within minutes.

② Broad Network Access

- Traditional on-premise systems are generally limited to the organization's internal network whereas cloud services support remote access from any location.

① Resource Pooling

In on premise environments, hardware is dedicated to a single organization, leading to lower resource utilization and higher infrastructure costs.

② Rapid Elasticity

Traditional infrastructure requires purchasing additional hardware for increased demand, which is expensive and time-consuming. cloud computing provides instant scalability without additional hardware investment.

- ② The evolution of computer hardware laid the foundation for cloud computing by continuously improving processing power, storage capacity and networking capabilities

Hardware Evolution

① 1939, Berry brothers invented vacuum tube } 1st Gen

① 1941 Turing complete

① 1946 ENIAC General purpose Computer } 2nd Gen

① 1971 IC used in mini computer, Intel 4004 } 3rd Gen
world's 1st commercial microprocessor

① 1974 commercially available personal computer } 4th Gen
was MITS Altair 8800

Internet Software Evolution

① 1957, launch of Sputnik I led US to establish ARPA
which Advanced Research Project Agency

① SAGE semi-Automatic Ground Environment a project with 800 programmers

① ~~Interface Message Processor~~ Network control Protocol 1st protocol used by ARPANET

① TCP/IP by Lawrence Robert in April (1967)

① IPv6 (1995)

① 1st webserver by Tim-Berners Lee

① Mosaic Browser

① Mozilla (1994)

- 3 Server virtualization is a key technology that enables cloud computing. It divides a single physical server into multiple VMs using hypervisor. Each virtual machine runs its own operating system and application independently. This improves resources utilization, reduces costs, and provides the flexibility required for cloud services.

Role of Server Virtualization in Cloud

- ① Efficient Resource Utilization
- ① Cost Reduction
- ① Rapid Provisioning
- ① Scalability
- ① Isolation and Security

Virtualization

- ① An enabling technology
- ① Creates VM using a hypervisor
- ① Focuses on hardware abstraction and resource sharing
- ① Improves server utilization

Cloud Computing

- ① A service delivery model
- ① Delivers computing services over the Internet
- ① Focuses on providing on-demand service to users
- ① Provides IaaS, PaaS, SaaS and XaaS service.

Data center

It is a facility that houses the physical infrastructure required for cloud computing, including servers, storage system, networking devices, power supplies and cooling system. Cloud service providers such as Amazon Web Services, Microsoft Azure and Google Cloud operate large data centers to deliver reliable and scalable cloud service to users worldwide.

Role of Data Centers in Cloud services

- ① Computing Resource
- ② Storage Services
- ③ Network Connectivity
- ④ Scalability
- ⑤ Reliability and ~~High~~ High Availability
- ⑥ Security

⑤ IaaS

- Infrastructure as a Service
- Provides virtual servers, storage and networking
- Manages hardware virtualization and networking
- Manages OS, application and data
- Eg Amazon EC2 (AWS)

PaaS

- Platform as a Service
- Provides platform to develop and deploy application
- Manages infrastructure, OS, runtime and development tools
- Develops and manages application and data
- Eg Google App Engine

SaaS

- Software as a Service
- Provides ready to use software through the Internet
- Manages the entire application and infrastructure
- Uses the software
- Eg Google Workspace (Gmail)

Cloud

- ① Anything as a service
- ② Provides various IT services such as storage, security or database over the cloud.
- ③ Manages the complete service and infrastructure
- ④ Uses the required service
- ⑤ Eg Dropbox (Storage as a service)